INCH-POUND

MIL-PRF-39003/6D
16 December 2003
SUPERSEDING
MIL-PRF-39003/6C
17 May 1996

PERFORMANCE SPECIFICATION SHEET

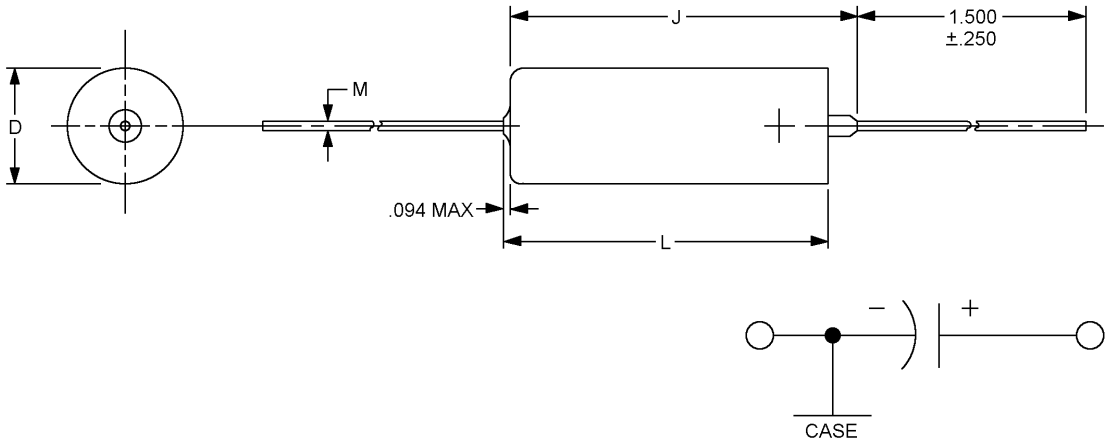
CAPACITOR, FIXED, ELECTROLYTIC (SOLID ELECTROLYTE), TANTALUM (POLARIZED, SINTERED SLUG), ESTABLISHED RELIABILITY, STYLE CSR33

FAILURE RATE LEVELS “M”, “P”, AND “R” ARE INACTIVE FOR NEW DESIGN AFTER 2 FEBRUARY 1984.

FAILURE RATE LEVEL “S” IS INACTIVE FOR NEW DESIGN AFTER 15 AUGUST 1984.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-39003.



CIRCUIT DIAGRAM

Inches	mm
.094	2.39
.250	6.35
1.500	38.10

CIRCUIT DIAGRAM

FIGURE 1. Dimensions and configuration.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Case size	Dimensions			
	L ± .031 (0.79)	D + .016 (0.41) - .015 (0.38)	M ± .002 (0.05)	J max
A	.286 (7.26)	.135 (3.43)	.020 (0.51)	.422 (10.72)
B	.474 (12.04)	.185 (4.70)	.020 (0.51)	.610 (15.49)
C	.686 (17.42)	.289 (7.34)	.025 (0.64)	.822 (20.88)
D	.786 (19.96)	.351 (8.92)	.025 (0.64)	.922 (23.42)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parentheses and are given for general information only.
3. The case insulation shall extend .015 (0.38 mm) minimum beyond each end. However, when a shrink-fitted insulation is used, it shall lap over the ends of the capacitor body.
4. Lead length may be a minimum of 1-inch long for use in tape and reel automatic insertion equipment, when specified.

FIGURE 1. Dimensions and configuration – Continued.

REQUIREMENTS:

Requirements: Requirements shall be in accordance with MIL-PRF-39003, and as specified herein.

Design and construction: Capacitors shall be of the design, construction, and physical dimensions specified on figure 1.

Case type: Tubular, insulated.

Material: Metal (alloy unspecified).

Type of seal: Hermetic.

Leads: Type N-32, N-52, W-32 or W-52 of MIL-STD-1276.

DC rated voltage: See table I. Above +85°C voltage derating is required (see MIL-PRF-39003).

DC leakage (DCL): See table I.

Capacitance (Cap.) (nom): See table I.

Cap. tolerance: See table I.

Dissipation factor (DF): See table I.

Operating temperature: -55°C to +125°C.

Failure rate level: See table I.

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TABLE I. Electrical characteristics and PIN's.

DC rated voltage	Capacitance	Capacitance tolerance	DC leakage at		Dissipation factor at -55°C through +125°C	Case size	PIN M39003/06- 1/ failure rate level for (%1000 hrs.)						
			+25°C	+85°C +125°C			M 1.0	P 0.1	R 0.01	S 0.001	B 0.1	C 0.01	D 0.001
volts	µF	percent	µA	µA	percent								
6	10	10	0.5	2	6	A	0001-	0101-	0201-	0301-	2001-	3001-	4001-
6	10	20	0.5	2	6	A	0002-	0102-	0202-	0302-	2002-	3002-	4002-
6	12	10	0.5	2	6	A	0003-	0103-	0203-	0303-	2003-	3003-	4003-
6	100	10	1	3	8	B	0004-	0104-	0204-	0304-	2004-	3004-	4004-
6	100	20	1	3	8	B	0005-	0105-	0205-	0305-	2005-	3005-	4005-
6	330	10	2	8	8	C	0006-	0106-	0206-	0306-	2006-	3006-	4006-
6	330	20	2	8	8	C	0007-	0107-	0207-	0307-	2007-	3007-	4007-
6	390	10	2	8	10	C	0008-	0108-	0208-	0308-	2008-	3008-	4008-
6	470	10	2	8	10	C	0009-	0109-	0209-	0309-	2009-	3009-	4009-
6	470	20	2	8	10	C	0010-	0110-	0210-	0310-	2010-	3010-	4010-
6	680	10	5	10	10	D	0011-	0111-	0211-	0311-	2011-	3011-	4011-
6	680	20	5	10	10	D	0012-	0112-	0212-	0312-	2012-	3012-	4012-
6	820	10	5	10	10	D	0013-	0113-	0213-	0313-	2013-	3013-	4013-
6	1000	10	5	10	10	D	0014-	0114-	0214-	0314-	2014-	3014-	4014-
6	1000	20	5	10	10	D	0015-	0115-	0215-	0315-	2015-	3015-	4015-
10	6.8	10	0.5	2	6	A	0016-	0116-	0216-	0316-	2016-	3016-	4016-
10	6.8	20	0.5	2	6	A	0017-	0117-	0217-	0317-	2017-	3017-	4017-
10	8.2	10	0.5	2	6	A	0018-	0118-	0218-	0318-	2018-	3018-	4018-
10	47	10	1	2	6	B	0019-	0119-	0219-	0319-	2019-	3019-	4019-
10	47	20	1	2	6	B	0020-	0120-	0220-	0320-	2020-	3020-	4020-
10	56	10	1	4	6	B	0021-	0121-	0221-	0321-	2021-	3021-	4021-
10	68	10	1	4	6	B	0022-	0122-	0222-	0322-	2022-	3022-	4022-
10	68	20	1	4	6	B	0023-	0123-	0223-	0323-	2023-	3023-	4023-
10	82	10	1	4	6	B	0024-	0124-	0224-	0324-	2024-	3024-	4024-
10	220	10	1	7	8	C	0025-	0125-	0225-	0325-	2025-	3025-	4025-
10	220	20	1	7	8	C	0026-	0126-	0226-	0326-	2026-	3026-	4026-
10	270	10	2	10	8	C	0027-	0127-	0227-	0327-	2027-	3027-	4027-
10	390	10	2	16	10	D	0028-	0128-	0228-	0328-	2028-	3028-	4028-
10	470	10	4	16	10	D	0029-	0129-	0229-	0329-	2029-	3029-	4029-
10	470	20	4	16	10	D	0030-	0130-	0230-	0330-	2030-	3030-	4030-
10	560	10	4	16	10	D	0031-	0131-	0231-	0331-	2031-	3031-	4031-

See footnote at end of table.

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TABLE I. Electrical characteristics and PIN's. – Continued.

DC rated voltage	Capacitance	Capacitance tolerance	DC leakage at		Dissipation factor at -55°C through +125°C	Case size	PIN M39003/06- 1/ failure rate level for (%1000 hrs.)						
			+25°C	+85°C +125°C			M 1.0	P 0.1	R 0.01	S 0.001	B 0.1	C 0.01	D 0.001
volts	µF	percent	µA	µA	percent								
15	4.7	10	0.5	2	4	A	0032-	0132-	0232-	0332-	2032-	3032-	4032-
15	4.7	20	0.5	2	4	A	0033-	0133-	0233-	0333-	2033-	3033-	4033-
15	5.6	10	0.5	2	4	A	0034-	0134-	0234-	0334-	2034-	3034-	4034-
15	33	10	1	2	6	B	0035-	0135-	0235-	0335-	2035-	3035-	4035-
15	33	20	1	2	6	B	0036-	0136-	0236-	0336-	2036-	3036-	4036-
15	39	10	1	2	6	B	0037-	0137-	0237-	0337-	2037-	3037-	4037-
15	150	10	1	7	8	C	0038-	0138-	0238-	0338-	2038-	3038-	4038-
15	150	20	1	7	8	C	0039-	0139-	0239-	0339-	2039-	3039-	4039-
15	180	10	2	10	8	C	0040-	0140-	0240-	0340-	2040-	3040-	4040-
15	220	10	2	10	8	D	0041-	0141-	0241-	0341-	2041-	3041-	4041-
15	220	20	2	10	8	D	0042-	0142-	0242-	0342-	2042-	3042-	4042-
15	270	10	2	16	8	D	0043-	0143-	0243-	0343-	2043-	3043-	4043-
15	330	10	2	16	8	D	0044-	0144-	0244-	0344-	2044-	3044-	4044-
15	330	20	2	16	8	D	0045-	0145-	0245-	0345-	2045-	3045-	4045-
20	2.7	10	0.5	2	4	A	0046-	0146-	0246-	0346-	2046-	3046-	4046-
20	3.3	10	0.5	2	4	A	0047-	0147-	0247-	0347-	2047-	3047-	4047-
20	3.3	20	0.5	2	4	A	0048-	0148-	0248-	0348-	2048-	3048-	4048-
20	3.9	10	0.5	2	4	A	0049-	0149-	0249-	0349-	2049-	3049-	4049-
20	18	10	1	2	6	B	0050-	0150-	0250-	0350-	2050-	3050-	4050-
20	22	10	1	2	6	B	0051-	0151-	0251-	0351-	2051-	3051-	4051-
20	22	20	1	2	6	B	0052-	0152-	0252-	0352-	2052-	3052-	4052-
20	27	10	1	2	6	B	0053-	0153-	0253-	0353-	2053-	3053-	4053-
20	56	10	1	10	6	C	0054-	0154-	0254-	0354-	2054-	3054-	4054-
20	68	10	1	10	6	C	0055-	0155-	0255-	0355-	2055-	3055-	4055-
20	68	20	1	10	6	C	0056-	0156-	0256-	0356-	2056-	3056-	4056-
20	82	10	1	10	6	C	0057-	0157-	0257-	0357-	2057-	3057-	4057-
20	100	10	1	10	6	C	0058-	0158-	0258-	0358-	2058-	3058-	4058-
20	100	20	1	10	6	C	0059-	0159-	0259-	0359-	2059-	3059-	4059-
20	120	10	1	10	6	C	0060-	0160-	0260-	0360-	2060-	3060-	4060-
20	150	10	2	10	8	D	0061-	0161-	0261-	0361-	2061-	3061-	4061-
20	150	20	2	10	8	D	0062-	0162-	0262-	0362-	2062-	3062-	4062-
20	180	10	2	10	8	D	0063-	0163-	0263-	0363-	2063-	3063-	4063-

See footnote at end of table.

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TABLE I. Electrical characteristics and PIN's. – Continued.

DC rated voltage	Capacitance	Capacitance tolerance	DC leakage at		Dissipation factor at -55°C through +125°C	Case size	PIN M39003/06- 1/ failure rate level for (%1000 hrs.)						
			+25°C	+85°C +125°C			M 1.0	P 0.1	R 0.01	S 0.001	B 0.1	C 0.01	D 0.001
volts	µF	percent	µA	µA	percent								
35	1.8	10	0.5	2	4	A	0064-	0164-	0264-	0364-	2064-	3064-	4064-
35	8.2	10	1	2	6	B	0065-	0165-	0265-	0365-	2065-	3065-	4065-
35	10	10	1	2	6	B	0066-	0166-	0266-	0366-	2066-	3066-	4066-
35	10	20	1	2	6	B	0067-	0167-	0267-	0367-	2067-	3067-	4067-
35	33	10	1	5	6	C	0068-	0168-	0268-	0368-	2068-	3068-	4068-
35	33	20	1	5	6	C	0069-	0169-	0269-	0369-	2069-	3069-	4069-
35	39	10	1	5	6	C	0070-	0170-	0270-	0370-	2070-	3070-	4070-
35	47	10	1	5	6	C	0071-	0171-	0271-	0371-	2071-	3071-	4071-
35	47	20	1	5	6	C	0072-	0172-	0272-	0372-	2072-	3072-	4072-
35	56	10	2	10	6	D	0073-	0173-	0273-	0373-	2073-	3073-	4073-
35	68	10	2	10	6	D	0074-	0174-	0274-	0374-	2074-	3074-	4074-
35	68	20	2	10	6	D	0075-	0175-	0275-	0375-	2075-	3075-	4075-
50	1.2	10	0.5	2	4	A	0076-	0176-	0276-	0376-	2076-	3076-	4076-
50	1.5	10	0.5	2	4	A	0077-	0177-	0277-	0377-	2077-	3077-	4077-
50	1.5	20	0.5	2	4	A	0078-	0178-	0278-	0378-	2078-	3078-	4078-
50	5.6	10	1	2	4	B	0079-	0179-	0279-	0379-	2079-	3079-	4079-
50	6.8	10	1	2	6	B	0080-	0180-	0280-	0380-	2080-	3080-	4080-
50	6.8	20	1	2	6	B	0081-	0181-	0281-	0381-	2081-	3081-	4081-
50	22	10	1	5	6	C	0082-	0182-	0282-	0382-	2082-	3082-	4082-
50	22	20	1	5	6	C	0083-	0183-	0283-	0383-	2083-	3083-	4083-
50	27	10	1	5	6	C	0084-	0184-	0284-	0384-	2084-	3084-	4084-
50	33	10	1	9	6	D	0085-	0185-	0285-	0385-	2085-	3085-	4085-
50	33	20	1	9	6	D	0086-	0186-	0286-	0386-	2086-	3086-	4086-
50	39	10	1	9	6	D	0087-	0187-	0287-	0387-	2087-	3087-	4087-

1/ Dash number shall include the letter "H" to indicate an internal construction solder with a minimum melting point of 221°C, or the "H" shall be deleted.

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REQUIREMENTS – Continued:

Stability at low and high temperatures: In accordance with MIL-PRF-39003.

Step 1 (+25°C):

DCL: See table I.

Cap.: Within tolerance of table I value.

DF: See table I.

Step 2 (-55°C):

ΔCap.: Shall not change more than ± 10 percent of step 1 measured value.

DF: See table I.

Step 3 (+25°C):

DCL: See table I.

ΔCap.: Shall not change more than ± 2 percent of step 1 measured value.

DF: See table I.

Step 4 (+85°C):

DCL: See table I.

ΔCap.: Shall not change more than ± 8 percent of step 1 measured value.

DF: See table I.

Step 5 (+125°C):

DCL: See table I.

ΔCap.: Shall not change more than ± 12 percent of step 1 measured value.

DF: See table I.

Step 6 (+25°C):

DCL: See table I.

ΔCap.: Shall not change more than ± 2 percent of step 1 measured value.

DF: See table I.

Surge voltage: In accordance with MIL-PRF-39003.

DCL: See table I.

ΔCap.: Shall not change more than ± 2 percent of initial measured value.

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DF: See table I.

Surge current:: This test shall be performed following thermal shock and voltage aging on initial qualification and on each group A inspection lot thereafter. A 5 percent defective allowed shall apply for group A for FRL's P, R, and S. Capacitors shall be subjected to 3 consecutive cycles of the conditions specified as follows:

Temperature: +25°C ± 5°C.

Apply rated voltage ± 2 percent for 4 ± 1 seconds.

Discharge the capacitors for 4 ± 1 seconds to a voltage below 1 percent of rated voltage.

The total dc resistance (excluding the capacitor) including the wiring, fixturing and output impedance of the regulated power supply to each test position during the charging cycle shall be 1.0 ± 0.2 ohms.

On the completion of the test, the capacitors shall meet the following requirements:

DCL: See table I.

ΔCap.: Shall be as specified in table I.

DF: See table I.

Life: Method 108 of ML-STD-202.

240, 1,000, and 2,000 hours:

At +25°C (as applicable):

DCL: See table I.

ΔCap.: Shall not change more than ± 5 percent of initial measured value.

DF: See table I.

At +85°C:

DCL: See table I.

At +125°C:

DCL: See table I.

4,000, 6,000, 8,000, and 10,000 hours (exponential only):

At +25°C (as applicable):

DCL: 200 percent of initial limit.

ΔCap.: Shall not change more than ±10 percent of initial measured value.

DF: See table I.

At +85°C:

DCL: 200 percent of initial limit.

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At +125°C:

DCL: 200 percent of initial limit.

Marking: In accordance with MIL-PRF-39003.

Part or Identifying Number (PIN): M39003/06-(dash number from table I).

Substitutability data: Substitutability for failure rate levels and capacitance tolerances shall be in accordance with MIL-PRF-39003.

* NOTES:

Reference documents. In addition to MIL-PRF-39003, this specification sheet references the following documents.

DEPARTMENT OF DEFENSE

MIL-STD-202 - Test Method Standard, Electronic and Electrical Component Parts.

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch or www.dodssp.daps.mil or from the Standardization Document Order Desk, 700 Robbins Avenue 4D, Philadelphia, PA 19111-5094).

* Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5910-2203-02)

Review activities:
Navy - AS, MC, OS, SH
Air Force - 19, 99

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